

PECCI HRIS — Technical Documentation

PECCI Multipurpose Cooperative

Human Resource Information System

Version: 1.1.0 **Year:** 2026 **Developed by:** UST Interns — Arkin Reinier Aguilar, Maxenne De Guzman, Bernice Elyssa Soriano, Emily Etea

Repository: <https://github.com/EmilyEtea/PECCI-HRIS>

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1. System Overview

The PECCI HRIS (Human Resource Information System) is a web-based application developed for PECCI Multipurpose Cooperative (PECCI) to centralize and streamline HR operations. The system covers the full employee lifecycle from onboarding to payroll processing.

1.1 Purpose

- Centralize employee records management
- Automate attendance tracking and computation
- Streamline leave application and approval workflows
- Automate payroll computation with Philippine regulatory compliance
- Provide management reports and audit trails

1.2 Scope

The system covers:

- Employee information management (CRUD)
 - Department and position management
 - Attendance tracking (Time In/Out with grace period logic)
 - Leave management with approval workflow
 - Payroll computation (BIR TRAIN Law compliant)
 - Reports generation
 - User account management with role-based access
 - System settings management
 - Audit trail logging
-

2. Technology Stack

Component	Technology	Version
Framework	ASP.NET Core MVC	.NET 10.0
ORM	Entity Framework Core	9.0.4
Database	SQL Server / LocalDB	2019+
Authentication	Cookie-based (ASP.NET Core)	Built-in
Password Hashing	BCrypt.Net-Next	4.0.3
PDF Generation	iText7	8.0.4
Excel Export	ClosedXML	0.104.2
Frontend Framework	Bootstrap	5.3.3
Icons	Font Awesome	6.5.0
Charts	Chart.js	4.4.2
IDE	Visual Studio Code / Visual Studio 2022	—
Version Control	Git / GitHub	—

3. System Architecture

3.1 MVC Pattern

The system follows the Model-View-Controller (MVC) architectural pattern:

```
PECCI_HRIS/
├── Configuration/           # Strongly-typed settings
│   ├── AttendanceSettings.cs
│   ├── PayrollSettings.cs
│   └── KioskSettings.cs
├── Controllers/            # 14 MVC controllers
│   ├── AccountController   # Authentication (Login, Logout, Profile)
│   ├── DashboardController # Dashboard statistics
│   ├── EmployeeController  # Employee CRUD + Profile
│   ├── DepartmentController # Department management
│   ├── PositionController  # Position management
│   ├── AttendanceController # Time In/Out, Scanner, Adjust, Summary
│   ├── LeaveController     # Leave applications, credits, types
│   ├── DeductionController # Employee loan/other deductions
│   ├── PayrollController   # Payroll computation, payslips
│   ├── ReportsController   # Report generation
│   ├── UsersController     # User account management
│   ├── SettingsController  # System settings
│   ├── AuditLogController  # Audit trail viewer
│   └── BaseController      # Shared helper methods
├── Data/
│   └── ApplicationDbContext # EF Core DbContext + seed data
├── Models/                  # 14 entity models
├── Services/                # 4 business logic services
│   ├── AttendanceComputationService
│   ├── TaxComputationService
│   ├── LeaveCreditService
│   └── AuditService
├── ViewModels/              # View models for all forms & displays
├── Views/                   # 55+ Razor views
│   ├── Account/
│   ├── Attendance/         # Index, Scanner, Adjust, Summary
│   ├── Deduction/          # Index, Create, Edit
│   ├── Dashboard/
│   ├── Employee/
│   ├── Leave/
│   └── Payroll/
```

```
| | Reports/
| | Settings/
| | Shared/          # _Layout.cshtml, _ValidationScriptsPartial
| | wwwroot/
| |   | css/pecci-theme.css # PECCI brand CSS theme
| |   | images/pecci-logo.png
| |   | favicon.png       # Browser tab icon (PECCI logo)
```

3.2 Dependency Injection

All services are registered in `Program.cs`:

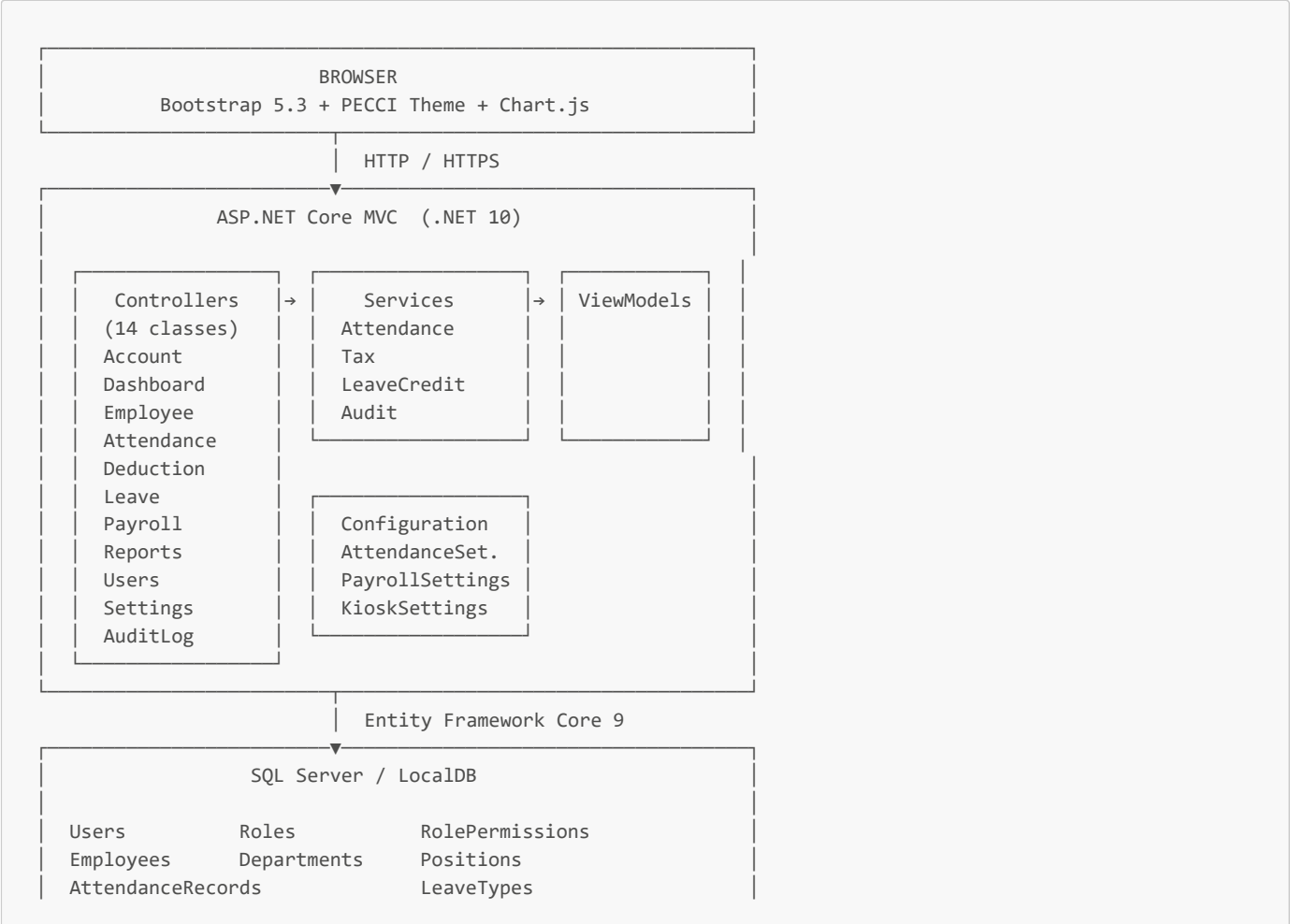
```
builder.Services.AddScoped<AttendanceComputationService>();
builder.Services.AddScoped<TaxComputationService>();
builder.Services.AddScoped<AuditService>();
builder.Services.AddScoped<LeaveCreditService>();
```

3.3 Configuration Binding

Settings are bound to strongly-typed classes:

```
builder.Services.Configure<AttendanceSettings>(
    builder.Configuration.GetSection("AttendanceSettings"));
builder.Services.Configure<PayrollSettings>(
    builder.Configuration.GetSection("PayrollSettings"));
builder.Services.Configure<KioskSettings>(
    builder.Configuration.GetSection("KioskSettings"));
```

3.4 System Architecture Diagram



LeaveApplications	LeaveCredits
EmployeeDeductions	PayrollRecords
AuditLogs	SystemSettings

3.5 Request Flow



4. Database Design

4.1 Entity Relationship Diagram



4.2 Table Descriptions

Users

Column	Type	Description
UserID	int PK	Auto-increment primary key

Column	Type	Description
Username	nvarchar(50) UNIQUE	Login username
PasswordHash	nvarchar(max)	BCrypt hashed password
Email	nvarchar(100) UNIQUE	User email
RoleID	int FK → Roles	Role assignment
EmployeeID	int FK → Employees nullable	Linked employee record
IsActive	bit	Account active/inactive
LastLogin	datetime2 nullable	Last successful login timestamp
CreatedAt	datetime2	Account creation timestamp

Employees

Column	Type	Description
EmployeeID	int PK	Auto-increment
EmployeeNo	nvarchar(20) UNIQUE	e.g., EMP-0001
FirstName	nvarchar(50)	First name
MiddleName	nvarchar(50) nullable	Middle name
LastName	nvarchar(50)	Last name
Suffix	nvarchar(10) nullable	Jr., Sr., III, etc.
DateOfBirth	datetime2	Date of birth
Gender	nvarchar(10)	Male / Female
CivilStatus	nvarchar(20) nullable	Single / Married / etc.
Nationality	nvarchar(20) nullable	Default: Filipino
Address	nvarchar(300) nullable	Home address
ContactNumber	nvarchar(20) nullable	Philippine mobile format
PersonalEmail	nvarchar(100) nullable	Personal email
CompanyEmail	nvarchar(100) nullable	Work email
SSSNumber	nvarchar(20) nullable	Format: XX-XXXXXXX-X
PhilHealthNumber	nvarchar(20) nullable	Format: XX-XXXXXXXXXX-X
PagIbigNumber	nvarchar(20) nullable	Format: XXXX-XXXX-XXXX
TINNumber	nvarchar(20) nullable	Format: XXX-XXX-XXX
DepartmentID	int FK → Departments	Department assignment
PositionID	int FK → Positions	Position assignment
DateHired	datetime2	Employment start date
DateRegularized	datetime2 nullable	Regularization date
DateSeparated	datetime2 nullable	Separation date
EmploymentStatus	nvarchar(30)	Regular / Probationary / Contractual / Part-time
Status	nvarchar(20)	Active / Inactive / Resigned / Terminated / Retired
CreatedAt	datetime2	Record creation timestamp
CreatedBy	int nullable	UserID who created the record

AttendanceRecords

Column	Type	Description
AttendanceID	int PK	Auto-increment
EmployeeID	int FK → Employees	Employee reference
AttendanceDate	datetime2	Date of attendance
TimeIn	time nullable	Time-in (HH:mm:ss)
TimeOut	time nullable	Time-out (HH:mm:ss)
BreakOut	time nullable	Break start
BreakIn	time nullable	Break end
LateMinutes	float nullable	Computed late minutes
OvertimeMinutes	float nullable	Computed overtime minutes
UndertimeMinutes	float nullable	Computed undertime minutes
TotalHoursWorked	float nullable	Computed total hours
AttendanceStatus	nvarchar(20)	Present / Late / Absent / On Leave / Holiday
IsManualEntry	bit	True if HR-adjusted
Remarks	nvarchar(300) nullable	Adjustment notes
AdjustedBy	int nullable	UserID who adjusted
AdjustedAt	datetime2 nullable	Adjustment timestamp
CreatedAt	datetime2	Record creation timestamp

LeaveApplications

Column	Type	Description
LeaveApplicationID	int PK	Auto-increment
EmployeeID	int FK → Employees	Applicant
LeaveTypeID	int FK → LeaveTypes	Type of leave
StartDate	datetime2	Leave start date
EndDate	datetime2	Leave end date
NumberOfDays	decimal(5,1)	Computed number of days
Reason	nvarchar(500) nullable	Reason for leave
Status	nvarchar(20)	Pending / Pending HR / Approved / Rejected / Cancelled
ManagerApproverID	int nullable	Manager who acted
ManagerApprovedAt	datetime2 nullable	Manager action timestamp
ManagerRemarks	nvarchar(300) nullable	Manager remarks
HRApproverID	int nullable	HR who acted
HRApprovedAt	datetime2 nullable	HR action timestamp
HRRemarks	nvarchar(300) nullable	HR remarks
AppliedAt	datetime2	Application submission timestamp

EmployeeDeductions (new)

Column	Type	Description
DeductionID	int PK	Auto-increment
EmployeeID	int FK → Employees	Employee reference
DeductionType	nvarchar(50)	SSS Loan / Pag-IBIG Loan / PECCI Loan / Cash Advance / Other
Description	nvarchar(200)	Deduction description / reference number
Amount	decimal(18,2)	Deduction amount in PHP
CutoffPeriod	nvarchar(10)	1-15 or 16-30
Month	int	Month (1–12)
Year	int	Year (e.g., 2026)
Status	nvarchar(20)	Active / Applied / Cancelled
CreatedAt	datetime2	Record creation timestamp
CreatedBy	int nullable	UserID who created the record

Status lifecycle: **Active** → payroll picks it up → **Applied**. HR can set to **Cancelled** before payroll runs.

PayrollRecords

Column	Type	Description
PayrollID	int PK	Auto-increment
EmployeeID	int FK → Employees	Employee reference
PayPeriod	nvarchar(20)	e.g., "2026-04-1-15"
PeriodStart	datetime2	Cutoff start date
PeriodEnd	datetime2	Cutoff end date
BasicSalary	decimal(18,2)	Semi-monthly basic pay
OvertimePay	decimal(18,2)	Computed OT pay
HolidayPay	decimal(18,2)	Holiday pay
NightDifferential	decimal(18,2)	Night differential pay
Allowances	decimal(18,2)	Allowances
OtherEarnings	decimal(18,2)	Other earnings
SSSContribution	decimal(18,2)	SSS employee share
PhilHealthContribution	decimal(18,2)	PhilHealth employee share
PagIbigContribution	decimal(18,2)	Pag-IBIG employee share
WithholdingTax	decimal(18,2)	BIR withholding tax
LateDeductions	decimal(18,2)	Late deduction amount
UndertimeDeductions	decimal(18,2)	Undertime deduction amount
OtherDeductions	decimal(18,2)	Absent deductions + custom deductions
StoredGrossPay	decimal(18,2)	Snapshot of gross pay at compute time
StoredTotalDeductions	decimal(18,2)	Snapshot of total deductions
StoredNetPay	decimal(18,2)	Snapshot of net pay
WorkingDays	int	Working days in period
DaysWorked	int	Days employee actually worked

Column	Type	Description
DaysAbsent	int	Days absent
TotalOvertimeHours	float	Total OT hours
TotalLateMinutes	float	Total late minutes
Status	nvarchar(20)	Draft / Finalized / Released
CreatedAt	datetime2	Computation timestamp
FinalizedAt	datetime2 nullable	Finalization timestamp

SystemSettings

Column	Type	Description
SettingID	int PK	Auto-increment
SettingKey	nvarchar(100)	Setting identifier
SettingValue	nvarchar(max)	Current value
SettingGroup	nvarchar(50)	Attendance / Payroll / Tax / General
DataType	nvarchar(20)	string / int / decimal / bool / time
AllowedValues	nvarchar(200) nullable	Comma-separated allowed values
Description	nvarchar(200) nullable	Human-readable description
IsEditable	bit	Whether editable from UI
UpdatedAt	datetime2	Last update timestamp
UpdatedBy	int nullable	UserID who last updated

5. Module Documentation

5.1 Authentication Module (Sprint 1)

Controller: `AccountController`

Action	Method	Description
Login (GET)	GET	Display login form
Login (POST)	POST	Authenticate user, create cookie session
Logout	POST	Sign out, clear session
Profile	GET	View current user profile
AccessDenied	GET	Access denied page

Authentication Flow:

1. User submits username + password
2. System looks up user by username (active only)
3. `BCrypt.Verify()` checks password against stored hash
4. On success: creates `ClaimsPrincipal` with role, name, `employeeID` claims
5. Cookie issued with 8-hour expiry (30 days if "Remember Me")
6. Failed attempts are logged to `AuditLog`

5.2 Employee Management Module (Sprint 1)

Controller: `EmployeeController`

Action	Method	Access	Description
Index	GET	All	List employees with search/filter
Profile	GET	All	View employee profile
Create (GET)	GET	HR Admin, HR Staff	Add employee form
Create (POST)	POST	HR Admin, HR Staff	Save new employee
Edit (GET)	GET	HR Admin, HR Staff	Edit employee form
Edit (POST)	POST	HR Admin, HR Staff	Update employee
Deactivate	POST	HR Admin	Deactivate employee
GetPositionsByDepartment	GET	All	AJAX — load positions by dept

Auto-features on Create:

- Employee number auto-generated if blank (EMP-XXXX format)
- Leave credits automatically allocated for current year (all active leave types)

5.3 Attendance Module (Sprint 2)

Controller: `AttendanceController` **Service:** `AttendanceComputationService`

Action	Method	Access	Description
Index	GET	All	Attendance list with filters
TimeIn	POST	All	Record time-in
TimeOut	POST	All	Record time-out
Adjust (GET)	GET	HR Admin, HR Staff	Manual adjustment form
Adjust (POST)	POST	HR Admin, HR Staff	Save adjustment + recompute
Summary	GET	HR Admin, HR Staff	Monthly summary report
Scanner	GET	Anonymous	Barcode/ID scanner kiosk terminal
Scan	POST	Anonymous	Process barcode scan (JSON API)

Computation Logic (`AttendanceComputationService`):

- Late threshold = $\text{WorkStart} + \text{GracePeriodMinutes} + \text{GracePeriodSeconds}$
- Late minutes = $\text{ceil}(\text{TimeIn} - \text{WorkStart})$ if $\text{TimeIn} > \text{LateThreshold}$
- Overtime = $\text{floor}(\text{TimeOut} - \text{WorkEnd})$ if $\text{TimeOut} > \text{WorkEnd}$ AND minutes $\geq \text{OvertimeThreshold}$
- Undertime = $\text{ceil}(\text{WorkEnd} - \text{TimeOut})$ if $\text{TimeOut} < \text{WorkEnd}$
- Hours worked = $(\text{TimeOut} - \text{TimeIn}) - \text{LunchBreakDuration}$ (if worked through lunch)

Scanner / Kiosk Terminal: The `/Attendance/Scanner` page is a standalone kiosk display (no login required) that accepts barcode or RFID scans. It supports:

- Lookup by **Employee No** (exact match)
- Lookup by **Display Name** ("FirstName LastName")
- Lookup by **Full Name** ("LastName, FirstName MiddleName")
- Partial name match (first or last name, if unique)
- **Double-scan confirmation** — first scan shows a pending confirmation, second scan within 10 seconds commits the Time In or Time Out

Scan Flow:

1st scan → "CONFIRM TIME IN – scan again within 10s"
 2nd scan → TIME IN recorded ✓
 3rd scan → "CONFIRM TIME OUT – scan again within 10s"
 4th scan → TIME OUT recorded ✓

5.4 Leave Management Module (Sprint 3)

Controller: `LeaveController`

Action	Method	Access	Description
Index	GET	All	Leave applications list
Apply (GET)	GET	All	Leave application form
Apply (POST)	POST	All	Submit leave application
Review (GET)	GET	HR Admin, HR Staff, Manager	Review application
Review (POST)	POST	HR Admin, HR Staff, Manager	Approve/Reject
Cancel	POST	All (own)	Cancel pending application
Credits	GET	All	View leave credits
Types	GET	HR Admin, HR Staff	Leave types list
CreateType	GET/POST	HR Admin	Add leave type

Approval Workflow:

1. Employee applies → Status: "Pending"
2. Manager reviews → Status: "Pending HR" (approved) or "Rejected"
3. HR reviews → Status: "Approved" or "Rejected"
4. On Approval: $UsedCredits += NumberOfDays$, $PendingCredits -= NumberOfDays$
5. On Rejection: $PendingCredits -= NumberOfDays$ (released)

5.5 Deductions Module (*new*)

Controller: `DeductionController`

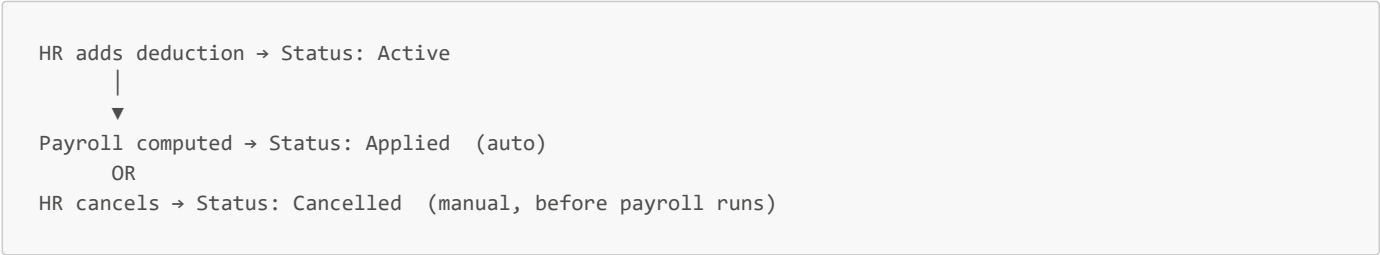
Action	Method	Access	Description
Index	GET	HR Admin, HR Staff	List deductions with filters
Create (GET)	GET	HR Admin, HR Staff	Add deduction form
Create (POST)	POST	HR Admin, HR Staff	Save new deduction
Edit (GET)	GET	HR Admin, HR Staff	Edit deduction form
Edit (POST)	POST	HR Admin, HR Staff	Update deduction
Cancel	POST	HR Admin, HR Staff	Cancel a deduction

Deduction Types:

- SSS Loan
- Pag-IBIG Loan
- PECCI Loan
- Cash Advance
- Other

Integration with Payroll: Deductions with `Status = "Active"` matching the employee, month, year, and cutoff period are automatically picked up during payroll computation and added to `OtherDeductions`. After payroll is computed, their status is set to `"Applied"`.

Status Lifecycle:



5.6 Payroll Module (Sprint 4)

Controller: PayrollController **Services:** TaxComputationService, AttendanceComputationService

Action	Method	Access	Description
Index	GET	HR Admin, HR Staff	Payroll records list
Compute (GET)	GET	HR Admin, HR Staff	Computation parameters form
Compute (POST)	POST	HR Admin, HR Staff	Run payroll computation
Payslips	GET	HR Admin, HR Staff	View payslips
Finalize	POST	HR Admin	Finalize payroll record

Full Computation Steps:

1. Determine period dates from cutoff selection (1–15 or 16–30)
2. Get employee's position `BasicSalary` ÷ 2 (semi-monthly)
3. Retrieve `AttendanceRecords` for the period
4. Compute: overtime pay, late deductions, **undertime deductions**, absent deductions
5. Fetch `EmployeeDeductions` (Active, matching month/year/cutoff) → sum as custom deductions
6. Compute government contributions (SSS, PhilHealth, Pag-IBIG) on monthly basis ÷ 2
7. Compute withholding tax on monthly taxable income ÷ 2
8. Store `StoredGrossPay`, `StoredTotalDeductions`, `StoredNetPay` as snapshots
9. Mark fetched `EmployeeDeductions` as "Applied"

Payroll Computation Flow:



6. Business Rules & Computations

6.1 Attendance Rules

Grace Period

```
LateThreshold = WorkStartTime + GracePeriodMinutes + GracePeriodSeconds
```

```
Default: 08:00 + 5 min + 0 sec = 08:05:00
```

```
If TimeIn > LateThreshold → LATE
```

```
LateMinutes = ceil(TimeIn - WorkStartTime) ← counted from 08:00, NOT from 08:05
```

Example:

- Time In: 08:06 → Late by 6 minutes (from 08:00)
- Time In: 08:05:00 → On time
- Time In: 08:05:01 → Late by 5 minutes (from 08:00)

Overtime

```
If TimeOut > WorkEndTime:
```

```
    OvertimeMinutes = floor(TimeOut - WorkEndTime)
```

```
    If OvertimeMinutes >= OvertimeThresholdMinutes → credited
```

```
    Else → not credited (below threshold)
```

Deductions

```
Daily Rate = MonthlySalary / 22 working days
```

```
Per-Minute Rate = Daily Rate / WorkingMinutesPerDay
```

```
Late Deduction = LateMinutes × Per-Minute Rate
```

```
Undertime Deduction = UndertimeMinutes × Per-Minute Rate
```

```
Absent Deduction = DaysAbsent × Daily Rate
```

6.2 BIR TRAIN Law Tax Computation (RA 10963, RR 2-2023)

Annual Tax Brackets (effective January 1, 2023):

Annual Taxable Income	Tax Formula
≤ ₱250,000	₱0 (Exempt)
₱250,001 – ₱400,000	(Income - 250,000) × 15%
₱400,001 – ₱800,000	₱22,500 + (Income - 400,000) × 20%
₱800,001 – ₱2,000,000	₱102,500 + (Income - 800,000) × 25%
₱2,000,001 – ₱8,000,000	₱402,500 + (Income - 2,000,000) × 30%
> ₱8,000,000	₱2,202,500 + (Income - 8,000,000) × 35%

Monthly Tax = Annual Tax / 12

Taxable Income = Gross Income - SSS - PhilHealth - Pag-IBIG

6.3 SSS Contribution (2026)

Employee share = 4.5% of Monthly Salary Credit (MSC) MSC range: ₱4,000 – ₱30,000 (stepped brackets)

6.4 PhilHealth Contribution (2026)

Total premium = Basic Salary × 5% Employee share = Total Premium / 2 (2.5%) Minimum: ₱500/month | Maximum: ₱5,000/month

6.5 Pag-IBIG Contribution

- Salary $\leq$ ₱1,500: Employee rate = 1%
- Salary $>$ ₱1,500: Employee rate = 2%
- Maximum employee contribution: ₱100/month

6.6 Overtime Pay Rates (DOLE)

Type	Multiplier
Regular overtime	1.25× hourly rate
Rest day overtime	1.30× hourly rate
Special holiday	1.30× hourly rate
Regular holiday	2.00× hourly rate
Night differential	+10% of hourly rate (10pm–6am)

7. Security Implementation

7.1 Authentication

- Cookie-based authentication (ASP.NET Core Identity-free)
- Session expires after 8 hours (sliding expiration)
- "Remember Me" extends to 30 days
- HttpOnly cookies prevent JavaScript access
- SecurePolicy = SameAsRequest (HTTPS in production)

7.2 Password Security

- BCrypt hashing with default cost factor (11 rounds)
- Passwords never stored in plain text
- Minimum 8 characters enforced on creation/reset

7.3 Authorization

- Role-based authorization via `[Authorize(Roles = "...")]`
- Four roles: HR Admin, HR Staff, Manager, Employee
- Employees can only view their own records
- HR Admin has full system access

7.4 Input Validation

- Server-side validation via Data Annotations
- Anti-forgery tokens on all POST forms (`@Html.AntiForgeryToken()`)
- EF Core parameterized queries prevent SQL injection
- Model binding prevents over-posting

7.5 Audit Trail

All user actions are logged to `AuditLogs` table:

- UserID, Username, Action, Module, Description
- IP Address, Timestamp
- Old values and new values (for updates)

8. Configuration & Settings

8.1 Connection String

File: `appsettings.json`

```
"ConnectionStrings": {  
  "DefaultConnection": "Server=(localdb)\\MSSQLLocalDB;Database=PECCI_HRIS_DB;..."
```

```
}
```

8.2 Adjustable Settings

All settings stored in `SystemSettings` table and editable via Admin → Settings UI.

Attendance Settings:

Key	Default	Description
WorkStartTime	08:00	Official work start
WorkEndTime	17:00	Official work end
GracePeriodMinutes	5	Grace period (08:00 + 5 = 08:05 cutoff)
GracePeriodSeconds	0	Additional seconds
OvertimeThresholdMinutes	30	Min OT before pay credited
LateDeductionType	PerMinute	PerMinute/PerHour/FixedAmount/None
LateDeductionAmountPerMinute	0	0 = auto from daily rate

Payroll Settings:

Key	Default	Description
CutoffDay1	15	First cutoff day
CutoffDay2	30	Second cutoff day
OvertimeRateMultiplier	1.25	Regular OT rate
RegularHolidayRateMultiplier	2.00	Holiday rate
NightDifferentialRate	0.10	Night diff rate

Tax Settings:

Key	Default	Description
TaxTableType	BIR_TRAIN_LAW_2023	Tax table to use
SSSEmployeeRate	0.045	SSS employee share
PhilHealthRate	0.025	PhilHealth employee share
PagIbigRate	0.02	Pag-IBIG rate
PagIbigMaxContribution	100.00	Max Pag-IBIG contribution

9. Installation Guide

9.1 Prerequisites

- .NET 10.0 SDK
- SQL Server 2019+ or LocalDB
- Visual Studio Code or Visual Studio 2022
- SQL Server Management Studio (SSMS) — optional

9.2 Setup Steps

Step 1: Clone Repository

```
git clone https://github.com/EmilyEtea/PECCI-HRIS.git
cd PECCI-HRIS/PECCI_HRIS
```

Step 2: Configure Connection String Edit appsettings.json:

```
"DefaultConnection": "Server=
(localdb)\\MSSQLLocalDB;Database=PECCI_HRIS_DB;Trusted_Connection=True;TrustServerCertificate=True;"
```

Step 3: Restore Packages

```
dotnet restore
```

Step 4: Run Migrations

```
dotnet ef migrations add InitialCreate
dotnet ef database update
```

Step 5: Run Application

```
dotnet run
```

Step 6: Login

- URL: https://localhost:5001
- Username: admin
- Password: Admin@123

9.3 Common Connection Strings

Scenario	Connection String
LocalDB	Server= (localdb)\\MSSQLLocalDB;Database=PECCI_HRIS_DB;Trusted_Connection=True;TrustServerCertificate=True;
SQL Express	Server=localhost\\SQLEXPRESS;Database=PECCI_HRIS_DB;Trusted_Connection=True;TrustServerCertificate=True;
SQL Auth	Server=localhost;Database=PECCI_HRIS_DB;User Id=sa;Password=YourPwd;TrustServerCertificate=True;

10. User Roles & Permissions

Module	HR Admin	HR Staff	Manager	Employee
Dashboard	✔ Full	✔ Full	✔ Full	✔ Own stats
Employee List	✔ Full	✔ Full	✔ View	✔ Own profile
Employee CRUD	✔ Full	✔ Create/Edit	✘	✘
Employee Deactivate	✔	✘	✘	✘
Departments	✔ Full	✔ Create/Edit	✘	✘
Positions	✔ Full	✔ Create/Edit	✘	✘
Attendance View	✔ All	✔ All	✔ View	✔ Own
Attendance Adjust	✔	✔	✘	✘
Scanner Terminal	✔	✔	✔	✔ (anonymous)
Leave Apply	✔	✔	✔	✔

Module	HR Admin	HR Staff	Manager	Employee
Leave Approve	✓	✓	✓ (Manager step)	✗
Leave Types	✓ Full	✓ View	✗	✗
Deductions	✓ Full	✓ Full	✗	✗
Payroll Compute	✓	✓	✗	✗
Payroll Finalize	✓	✗	✗	✗
Payslips	✓	✓	✗	✗
Reports	✓ All	✓ All	✓ All	✗
User Management	✓	✗	✗	✗
System Settings	✓	✗	✗	✗
Audit Trail	✓	✗	✗	✗

11. API Reference

AJAX / JSON Endpoints

GET /Employee/GetPositionsByDepartment

Returns positions for a given department (used in employee form dropdowns).

Parameters: `departmentId` (int)

Response:

```
[
  { "positionID": 1, "positionTitle": "HR Manager", "basicSalary": 65000.00 }
]
```

POST /Attendance/Scan

Processes a barcode or RFID scan from the kiosk terminal. Anonymous — no login required.

Request Body:

```
{ "employeeNo": "EMP-0001", "deviceIP": "192.168.1.10" }
```

`employeeNo` can be an Employee No, Display Name ("Juan Dela Cruz"), or Full Name ("Dela Cruz, Juan").

Response — Pending (first scan):

```
{
  "success": true,
  "action": "CONFIRM TIME IN",
  "employeeNo": "EMP-0001",
  "employeeName": "Juan Dela Cruz",
  "department": "Human Resources",
  "timeRecorded": "08:02:15",
  "message": "Scan again within 10s to confirm TIME IN.",
  "isLate": false,
  "isPending": true
}
```


Response — Confirmed (second scan):

```
{
  "success": true,
  "action": "TIME IN",
  "employeeNo": "EMP-0001",
  "employeeName": "Juan Dela Cruz",
  "department": "Human Resources",
  "timeRecorded": "08:02:18",
  "message": "On time",
  "isLate": false,
  "isPending": false
}
```

POST /Settings/UpdateSetting

Updates a single system setting. Requires HR Admin role.

Request Body:

```
{ "settingID": 3, "settingKey": "GracePeriodMinutes", "settingValue": "10" }
```

Response:

```
{ "success": true, "message": "GracePeriodMinutes updated successfully." }
```

GET /Settings/PreviewAttendanceRule

Returns computed late threshold for live preview in the Settings UI.

Parameters: `workStart` (HH:mm), `workEnd` (HH:mm), `graceMins` (int), `graceSecs` (int), `otThreshold` (int)

Response:

```
{
  "success": true,
  "lateThreshold": "08:10:00",
  "description": "Employees can time-in up to 08:10:00. Any time-in after that is marked LATE."
}
```

12. Known Issues & Limitations

Implemented but not yet wired

1. **PDF Payslip Export** — iText7 is included as a dependency. Payslips can currently be printed via browser (`Ctrl+P`). A dedicated PDF download endpoint is planned.
2. **Excel Report Export** — ClosedXML is included. Reports can be printed via browser. Export-to-Excel buttons are planned for a future sprint.

Not yet implemented

3. **Email Notifications** — Leave approval/rejection notifications via email are not in this version.
4. **Biometric Device Integration** — Time In/Out is web-based. The Scanner Terminal supports barcode/RFID via keyboard-wedge scanners. Native biometric device SDK integration is out of scope.

5. **Holiday Calendar** — Regular and special Philippine holidays must be manually marked in attendance records. An automated holiday calendar (based on Proclamation list) is planned.
6. **Night Differential Auto-Computation** — Night differential rate is defined in settings but not automatically applied during payroll. Manual entry via **OtherEarnings** is required for now.
7. **Recurring Deductions** — Each deduction entry covers one cutoff period. Recurring loan deductions (e.g., monthly SSS loan amortization) must be re-entered each cutoff. A recurring deduction schedule feature is planned.

Performance notes

8. **Leave Credit Refresh on Startup** — `LeaveCreditService.RefreshAnnualCredits()` runs on every app startup. It is idempotent (only creates missing records) but may be slow if there are many employees. Consider moving to a scheduled background job in production.

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